

Understanding Spanning Tree Protocol (STP)

The Role of the Root Bridge & Loop Prevention Mechanics

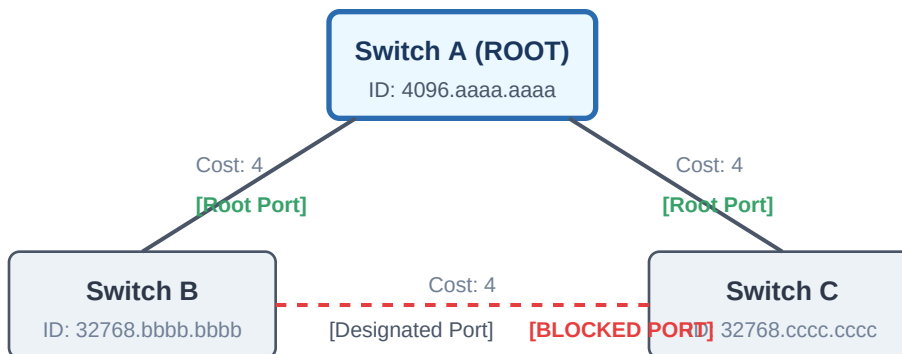
The Role of the STP Root Switch

- **Logical Reference Point:** Acts as the central hub and common denominator for the entire network topology.
- **BPDU Generation:** Originates Configuration Bridge Protocol Data Units (BPDUs) continuously to synchronize topology state.
- **Path Baseline:** Serves as the ultimate target destination for all shortest-path cost metrics across non-root switches.

How the Root Influences Link States

- **Root Path Cost:** Every non-root switch accumulates path costs of interfaces along the line back to the root.
- **Root Port (RP) Selection:** The single local port providing the cheapest mathematical cost back to the root is placed in Forwarding mode.
- **Designated Port (DP) Selection:** On any given link segment, the switch offering the lowest root path cost handles traffic forwarding for that segment.
- **Blocking Redundant Ports:** Interfaces that fail to win Root Port or Designated Port status are administratively Blocked to break traffic loops.

Visualizing the Three-Switch Triangle Topology



The Step-by-Step Decision Framework

When an STP topology converges, it runs through the following tie-breaking criteria in order:

Step Evaluation	Tie-Breaker Rule Applied	Outcome in Topology Above
1. Elect Root Bridge	Lowest Bridge ID (Priority + MAC Address)	Switch A wins with custom priority 4096.
2. Select Root Ports	Lowest calculated accumulated path cost back to Root	Switch B and Switch C both select their direct links (Cost 4).
3. Select Designated Ports	Segment owner with the lowest path cost to Root wins	On the shared B-C link, both have cost 4. Tied.
4. Final Link Tie-Breaker	Lowest sender Bridge ID determines designated status	Switch B's MAC address wins over C. Switch C's port is Blocked .